Wi-Fi CSI People Counting

Dataset audit findings, what is usable, and a step-by-step plan to an ESP32-S3 people counter

Prepared 2026-10-19 from the audit in D:/wifi/reports/dataset_audit/. Every claim below was checked against the files, the dataset's own documentation, or the official dataset page / paper. Items not yet verified are marked **(unverified)**.

Bottom line

- **Datasets with real people-count labels:** EHUNAM (0-8), WiMANS (0-5), the 80 MHz set in *Dataset/* (0 and 2-10), CountFi (1-8 and 10, very likely) and WiCount (0-3).
- **Only WiCount was recorded on an ESP32-S3**, and it has just 15 one-minute files covering 0-3 people. No public dataset combines ESP32-S3 hardware with a wide count range.
- **So the plan is:** learn what makes counting work from the large labelled sets, pre-train on CSI-Bench and the ESP32 sets, then **record your own ESP32-S3 data**. That last step decides real-world accuracy.
- **Related tasks only:** IIT-BHU (1-3 person activity scenarios), CSI HAR (empty vs one person), Haron98 (labels undefined) and CSI-Bench (no count labels at all).

1. What was audited

Seven folders in D:/wifi/Datasets were inspected file by file, read-only. Nothing was trained, resampled or normalised. For each dataset the audit established: what the files contain, every field and its meaning, the hardware and radio settings, how labels were assigned, signal quality and timing, and leakage risks. EHUNAM, the full WiMANS and CSI-Bench live in your Google Colab and have **not** been audited yet. Their entries below come from their papers or your survey PDF.

Dataset	Where	Audit depth
WiCount	D:/wifi/Datasets/WiCount	Every row of all 15 files
80 MHz Human Sensing (Dataset/)	D:/wifi/Datasets/Dataset	Every row of all 40 MAT files (streamed)
CountFi (Apr 2025)	D:/wifi/Datasets/CountFi	All 4.74 million packets decoded
IIT-BHU	D:/wifi/Datasets/IIT-BHU	Every line of all 928 files
CSI HAR	D:/wifi/Datasets/CSI HAR	Every row of all 10 sessions
Haron98	Colab (deleted locally)	Every line of all 4,800 files, plus one trial in detail
WiMANS (local partial copy)	Colab (deleted locally)	All 3,057 file headers + 77 files in full
EHUNAM, WiMANS (full), CSI-Bench	Colab	**Not audited yet** - Phase 0 of the plan

2. Findings, dataset by dataset

80 MHz Human Sensing - people-counting folders (Dataset/)

What it is	Meneghello et al., IEEE DataPort (doi 10.21227/xbhv-f125). Only the people-counting folders PC-1a to PC-4a are present.
Hardware / radio	Routers with Nexmon CSI firmware. 5 GHz, channel 42, 80 MHz, 802.11ac. One transmit stream, 4 receive antennas. 173 packets/s (published).
What the files hold	One variable per file, csi_buff: complex channel values, 256 frequency bins (242 usable). Rows cycle through the 4 antennas (verified). No timestamps, no signal-strength field.
Labels	File name nNN = number of people moving in the room: 0 and 2-10. **The 1-person files are missing** in all 4 setups, and PC2a_n00 is empty.
Watch out	One recording per (setup, count). Half the spectrum needs a documented sign flip. Setup descriptions (set_names.pdf) are not downloaded.

Verdict: the best-labelled counting data you have locally. Use it for method development and held-out-setup testing. It cannot be reproduced on an ESP32-S3 (5 GHz, 80 MHz, 4 antennas).

CountFi - bus passenger counting (CountFi/48)

What it is	Guo et al., WCNC 2025. Your folder is the **complete April 2025 release**: the page's download button links to *48.zip*. Licence CC BY-NC 4.0.
Hardware / radio	Two Raspberry Pi receivers with Nexmon. 5 GHz, channel 42, 80 MHz, single antenna (all from the files; the page gives no radio details). About 1,000 packets/s.
What the files hold	4,740 pcap files of 1,000 packets (~1 s) each. Each packet holds 256 complex values, signal strength, frame type and a microsecond timestamp. 10 files are empty and 4 hold only a header.
Labels	Numbered sub-folders 1-8 and 10. The page says every release ships with ground-truth counts and this download contains nothing else, so the folder number is **very likely the passenger count** (not stated in words).
Watch out	Two kinds of packet are mixed: data frames (45%, clean) and acknowledgement frames (53%, different structure); keep only data frames. Bins 0-1 and 125-131 hold junk values; dropping the standard 14 empty bins fixes it. The *head* and *end* receivers record the same seconds, and files are consecutive slices of a trip, so split by trip block, never by file.

Verdict: usable counting data (1-8, 10) once the labels are confirmed by the authors. Not ESP32-reproducible. The paper's 2023 set (*613_20People.zip*, counts up to 20) is worth downloading later.

WiCount (WiCount/)

What it is	Zhao et al., CSI-BERT2 (IEEE TMC 2025). Hugging Face RS2002/WiCount.
Hardware / radio	**ESP32-S3 receiver**, 1 antenna, 2.4 GHz, channel 12, 20 MHz, legacy frames. About 83 packets/s measured (100 claimed).
What the files hold	15 CSV files of 60 s. Each row holds 52 complex subcarriers (104 integers; empty slots already removed), signal strength and timestamps. 22 of 28 columns never change.
Labels	Folder = people count 0-3. Files per class: 1 / 7 / 6 / 1.
Watch out	Recorded in one evening in the order 0 -> 1 -> 2 -> 3 people, so count is tied to elapsed time. The README's I/Q order contradicts Espressif's documentation, and its process1.py has an indexing bug in the imaginary part.

Verdict: the only ESP32-S3 counting data. Too small to train on. Use it as the format reference and as an ESP32-S3 sanity test.

IIT-BHU multi-human activity (IIT-BHU/)

What it is	Gupta et al., IEEE DataPort (doi 10.21227/e6c6-aa21). Activity recognition with 1, 2 or 3 users.
Hardware / radio	ESP32 NodeMCU with ESP32-CSI-Tool. 2.4 GHz channel 6, mostly 40 MHz. Irregular 4-52 packets/s (median 28).
What the files hold	925 usable CSV files, 961k packets, 577 minutes. Each row: CSI (384 or 128 values), signal strength, timestamps.
Labels	Folder = number of users (1-3). File name = user IDs + activities (bend, clap, move, run, stretch, wave). **No empty-room class.**
Watch out	11 byte-identical file pairs under different user IDs; 3 empty files; the same users appear in several groups; about 2.5% of rows come from other transmitters or a different Wi-Fi mode.

Verdict: ESP32-format data for a 1-3 group-size task or pre-training, after removing duplicates and filtering rows.

CSI HAR (CSI HAR/)

What it is	Zhuravchak et al., figshare 14386892 (CC BY 4.0). One person, 3 rooms.
Hardware / radio	TP-Link TL-WDR4300 routers with Atheros CSI Tool. 5 GHz channel 60, 40 MHz, 2x2 antennas, 114 subcarriers.
What the files hold	67,449 packets. Each row: 114 constant index numbers + 4x114 signal strengths + 4x114 phases. The complex values can be recovered exactly. **No timestamps**, so the packet rate is unknown.
Labels	Per packet: walking, sitting, standing, lying, get up, get down, **no_person**. The label file is one row shorter than the data in every session.

Verdict: good for empty-room vs occupied detection and activity recognition. Cannot train a counter (maximum one person).

Haron98 - WiFi CSI Router Reception (now Colab only)

What it is	Hugging Face Haron98/WIFI-CSI-Dataset. 3 ESP32 receivers listening to one router.
What the files hold	1,600 trials x 3 receivers x 100 frames (~4 s, ~25 frames/s). 52 usable subcarriers per frame. Data is clean apart from 598 broken first lines.
Labels	**label_00-03 are undefined** (the README itself says TODO). label_00 looks quieter, but it was recorded last, so this can't define it.
Watch out	In the trial checked in detail, receiver dev3 was very weak (values within +/-9). The 3 receivers record the same moment, so keep them together when splitting.

Verdict: unlabelled ESP32 data. Becomes useful only if you learn what the labels mean.

Datasets in Colab - not yet audited

EHUNAM	0-8 people (big gap above 4), several rooms, Raspberry Pi with Nexmon plus Atheros, 2.4 and 5 GHz, .mat files with complex CSI, signal strength and timestamps. Each capture is saved once per receiver (3-4). **(Unverified; from your survey PDF.)**
WiMANS (full)	0-5 users (0: 594, 1: 3,564, 2-5: 1,782 samples each), 3 rooms, Intel 5300 3x3 antennas, 1 kHz, 3-s samples, labels in annotation.csv. The local copy had no labels, and 16 of its 31 sample groups showed uncorrelated signal over time (unexplained).
CSI-Bench	NeurIPS 2025, 461 hours, 26 rooms, 16 commodity devices **including ESP32-S3**. Signal strength only (no phase). Tasks: falls, breathing, location, motion source, activity, user ID, proximity. **No people-count labels.**

Verdict: EHUNAM and WiMANS are your main labelled training sets. CSI-Bench is for pre-training only. All three need the Phase 0 audit.

3. What is usable for what

Dataset	Counts available	ESP32-S3 match	Role in the plan
EHUNAM	0-8	No (Nexmon / Atheros)	Main labelled training
WiMANS (full)	0-5	No (Intel 5300)	Main labelled training
80 MHz (Dataset/)	0, 2-10	No (5 GHz, 4 antennas)	Labelled training + held-out-setup test
CountFi Apr 2025	1-8, 10 (very likely)	No (5 GHz, 80 MHz)	Labelled training (bus setting)
WiCount	0-3	**Yes**	ESP32-S3 format reference + sanity test
IIT-BHU	1-3 (no empty)	Mostly (ESP32)	ESP32 pre-training / 1-3 group size
CSI HAR	0-1 (presence)	No (Atheros)	Empty vs occupied stage
CSI-Bench	none	Partly (has ESP32-S3)	Self-supervised pre-training
Haron98	unknown	Yes (ESP32)	Unlabelled pre-training
Your own ESP32-S3 data	0-5+ (you choose)	**Yes**	**Fine-tuning and the final test**

Rules that apply to every dataset

- **Split by recording, setup, room, day or trip - never by random window.** Neighbouring windows from one recording are near-copies, so random splits give falsely high accuracy.
- **Keep simultaneous receivers on the same side of a split**: CountFi head/end, EHUNAM's receivers, Haron98 dev1-3.
- **Watch for count tied to time.** WiCount and Haron98 recorded classes in order, so a model can learn drift instead of occupancy. Record your own data in shuffled order.
- **Define windows in seconds, not packets.** Packet rates range from 25 per second to 1,000 per second. Mark lost packets as missing; don't invent them.
- **Drop columns that never change** and the empty subcarriers before modelling.

4. The plan

Six phases. Phases 0-3 run in Google Colab, because your D: drive has about 18 GB free and 16 GB of RAM. Phase 4 needs ESP32-S3 boards and a room. Rough durations assume part-time work.

Phase 0 - Close the gaps (1-2 days, Colab)

- [ ] Audit EHUNAM, the full WiMANS and CSI-Bench the same way as the local sets: shapes, subcarrier layout, packet rate, labels per recording, split units.
- [ ] WiMANS: build the 'users per sample' table from annotation.csv; explain the 16 groups with uncorrelated signal (probably band or room).
- [ ] EHUNAM: map every recording to its count using Summary.xlsx; group the 3-4 receiver copies of each capture.
- [ ] Email the CountFi authors to confirm that folder number = passenger count, and ask what '48', 'ms', 'v2' and 'v3' mean.
- [ ] Optional: download CountFi 613_20People.zip (paper's benchmark set, up to 20 people) and the 80 MHz set_names.pdf.

Phase 1 - One shared input format (2-3 days)

- [ ] Choose a format every dataset can be reduced to **and** an ESP32-S3 can produce: one link, signal strength only, about 52 subcarriers over a 20 MHz slice.
- [ ] For 80/40 MHz data, cut out one 20 MHz sub-band; for multi-antenna data, use one antenna at a time.
- [ ] Resample to one rate (e.g. 50 Hz) with 3-5 s windows; mark gaps as missing.
- [ ] Normalise per recording and per receiver (removes automatic gain changes and device differences).
- [ ] Also compute simple count cues per window: how much each subcarrier fluctuates, motion energy, and how many independent patterns the signal contains.

Phase 2 - Honest baselines per dataset (3-5 days)

- [ ] Train separately on each labelled dataset with held-out setups, rooms or days.
- [ ] Models: gradient boosting on the simple cues, and a small 1D-CNN / temporal ConvNet on the windows.
- [ ] Treat counting both as classification and as ordered regression.
- [ ] Report: average count error, accuracy within +/-1 person, per-count recall.
- [ ] Outcome: which count ranges are learnable and which features matter.

Phase 3 - Combine datasets (1-2 weeks)

- [ ] Pre-train one encoder without labels on everything in the shared format: CSI-Bench (especially its ESP32 parts), Haron98, IIT-BHU and the labelled sets.
- [ ] Add a count head trained on EHUNAM + WiMANS + 80 MHz + CountFi.
- [ ] Test by leaving one whole dataset out; try domain adaptation (removing dataset/device identity from the features).
- [ ] Check on WiCount: zero-shot first, then after a few minutes of fine-tuning.

Phase 4 - Your own ESP32-S3 recordings (1-2 weeks; decides real accuracy)

- [ ] Hardware: 1 ESP32-S3 sending fixed traffic at ~100 packets/s (e.g. ESP-NOW or ping) and 2-3 ESP32-S3 receivers, with the same firmware and logging format as WiCount.
- [ ] Counts 0-5 or more, recorded in **shuffled order**, several sessions per count, over several days and at least 2 rooms.
- [ ] Start every session with an empty-room recording.
- [ ] Include people sitting still and moving.
- [ ] Ground truth: a timestamped written log or a phone video. Note the room layout and device positions.
- [ ] Fine-tune the Phase 3 model; test on a day and a room it never saw.

Phase 5 - Run it on the ESP32-S3 (1 week)

- [ ] Model under ~100k parameters, quantised to 8-bit (TFLite Micro or ESP-DL), sized for the S3's 512 KB internal + 8 MB external memory.
- [ ] On-device processing identical to Phase 1: signal strength, normalise, 3-5 s window, plus a rolling empty-room baseline taken at installation.
- [ ] Two stages: a cheap presence detector first, then the counter. Smooth the output over time so it doesn't flicker.
- [ ] Measure speed, memory and accuracy on the device, against the Phase 4 test days.

5. What to do this week

- [ ] Open EHUNAM, WiMANS and CSI-Bench in Colab and run the Phase 0 audit notebooks (Claude can generate them).
- [ ] Send the CountFi authors the label question (contact: jingtao2023 dot guo at connect dot polyu dot hk).
- [ ] Order or set up 3-4 ESP32-S3 boards and flash the CSI receive / transmit firmware.
- [ ] Make one 10-minute test recording (empty room, then 1 person) and check it parses like WiCount.
- [ ] Decide the target count range for deployment (e.g. 0-5). This sets how much of Phase 4 you need.

6. Open questions

Question	Who / how to resolve
Are CountFi folder numbers the passenger counts? What are '48', 'ms', 'v2', 'v3'?	CountFi authors (email)
What do Haron98 label_00-03 mean?	Dataset uploader on Hugging Face
What are the 80 MHz setups PC-1a..4a? Why are the 1-person files absent?	set_names.pdf on IEEE DataPort
Which 52 subcarriers does WiCount keep, and in which I/Q order?	Compare with your own ESP32-S3 capture
What is CSI HAR's packet rate?	Authors' code (github.com/retsediv/WIFI_CSI_based_HAR)
EHUNAM counts, rooms and file layout	Phase 0 audit in Colab
Why do 16 WiMANS sample groups look uncorrelated over time?	annotation.csv (band / room) in Phase 0

7. Where everything is

File	Contents
D:/wifi/reports/dataset_audit/DATASET_AUDIT_REPORT.md	Full audit report
FEATURE_DICTIONARY.md	Every field of every dataset, with meaning and evidence
DATA_ACQUISITION_PROTOCOLS.md	Hardware, radio settings, rooms, labelling per dataset
DATASET_COMPARISON.md	Side-by-side comparison and which datasets can be combined
dataset_inventory.csv, label_distribution.csv, compatibility_matrix.csv	Tables
audit_findings.json, audit_errors.log	Machine-readable findings; problems found
scripts/ (01-12)	Re-runnable audit scripts: py -P -E scripts/<name>.py ...

Wiman/ and Haron98_WIFI-CSI-Dataset/ were deleted from D: on 2026-10-19 at your request (you use them in Colab). D: now has about 17.8 GB free.